

Lecture 5: Atmospheres I

Composition, Structure, & Energy Balance

Introduction to Planetary Science



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Learning Objectives

By the end of this lecture, you will be able to:

- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above T_{eff}
- Evaluate atmospheric escape mechanisms and retention criteria

Recap: Lecture 4

- Metal–silicate separation driven by magma oceans
- Hf–W chronometry: Earth's core formed in ~30–60 Myr
- Goldschmidt classification: siderophile, lithophile, chalcophile, atmophile
- Dynamo requirements: conducting fluid + convection + $R_m \gtrsim 10\text{--}100$
- Mars lost its dynamo ~4.1 Ga → atmospheric erosion

Today: The thin gaseous envelope that controls a planet's surface conditions.



1. Atmospheric Composition

Captured directly from the protoplanetary disk.

- Dominated by H_2 and He (solar composition)
- Requires mass $\gtrsim 5\text{--}10 M_{\oplus}$ to capture nebular gas
- Disk disperses in $\sim 3\text{--}10$ Myr (Lecture 2)
- **Gas giants:** Jupiter and Saturn, massive H_2/He envelopes
- **Ice giants:** Uranus and Neptune, envelopes only 10–20% of total mass
- Terrestrial planets too small $\rightarrow$ any primordial H quickly lost

Secondary Atmospheres

Produced by outgassing from the interior.

- Volcanism and magma ocean degassing release dissolved volatiles
- Speciation depends on **oxygen fugacity** (Lecture 4):
 - Oxidising: CO_2 , H_2O , N_2
 - Reducing: H_2 , CO , N_2
- **Venus:** CO_2 -dominated (96.5%), 92 bar
- **Mars:** CO_2 -dominated (95%), but only 6 mbar
- **Titan:** thick N_2 (from NH_3 conversion), 1.5 bar

Where Do Atmospheres Come From?

A pictorial summary of the three classes.

Primary Directly captured nebular gas; H_2/He dominated.
Requires $M \gtrsim 5\text{--}10 M_{\oplus}$ before disk dispersal.

Secondary Outgassed from the interior during accretion + cooling.
Composition set by f_{O_2} of the mantle.

Tertiary Modified by surface processes, photochemistry, and (for Earth) biology.

Over time, a planet's atmospheric inventory can be **replaced** or **re-equilibrated** entirely: the present composition may not reflect the initial inventory.

Tertiary Atmospheres

Modified by surface processes, photochemistry, or biology.

- Earth's original outgassed atmosphere: likely $\text{CO}_2 + \text{N}_2$ (similar to Venus)
- Oxygenic photosynthesis transformed it over billions of years
- Present atmosphere: N_2 (78%) + O_2 (21%)
- O_2 **is entirely biogenic**, and would disappear in ~Myr without life

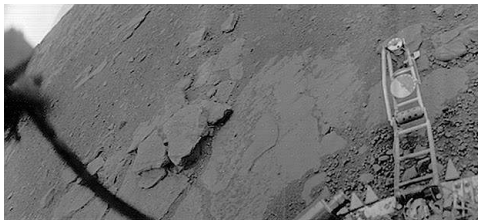
Earth's atmosphere is a **tertiary atmosphere**: the product of billions of years of biological modification.

Comparative Atmospheric Properties

	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	735	288	210	n/a	94
Dominant gas	CO ₂	N ₂	CO ₂	H ₂	N ₂
Secondary gas	N ₂	O ₂	N ₂	He	CH ₄
μ	43.4	29.0	43.3	2.2	28.6
Type	2 ⁺	3 ⁺	2 ⁺	1 ⁺	2 ⁺

Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 735 K.

Venus: A Secondary Atmosphere in Extremis



Credit: USSR/NASA NSSDC, public domain

- *Venera 13* lander, 1 March 1982
- Surface pressure: 92 bar
- Surface temperature: 735 K
- Orange sky: thick CO₂ atmosphere scatters light
- Lander survived 127 minutes

A vivid demonstration of how a massive secondary atmosphere transforms surface conditions.



2. Hydrostatic Equilibrium

Hydrostatic Equilibrium

Consider a thin slab of atmosphere at height z , thickness dz , area A :

- Pressure from below (upward): $P(z) \cdot A$
- Pressure from above (downward): $P(z + dz) \cdot A$
- Weight (downward): $\rho(z) g A dz$

Force balance:

$$P(z) A - P(z + dz) A = \rho g A dz$$

$$\frac{dP}{dz} = -\rho g$$

Pressure decreases with altitude: each layer supports the weight above it.

The Ideal Gas Law (Atmospheric Form)

To solve hydrostatic equilibrium, we need $\rho(P)$:

$$P = n k_B T = \frac{\rho k_B T}{\mu m_u}$$

n number density

k_B Boltzmann constant ($1.381 \times 10^{-23} \text{ J K}^{-1}$)

μ mean molecular weight (dimensionless, in amu)

m_u atomic mass unit ($1.661 \times 10^{-27} \text{ kg}$)

Rearranging: $\rho = P \mu m_u / (k_B T)$

The Barometric Formula

Substitute $\rho = P\mu m_u / (k_B T)$ into hydrostatic equilibrium:

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P$$

For an **isothermal** atmosphere (constant T , constant g), this separates:

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right)$$

where $H = k_B T / (\mu m_u g)$ is the **pressure scale height**.

- Pressure drops by $e \approx 2.718$ every scale height
- 99% of the atmospheric mass lies below $\sim 5H$

3. Blackboard Derivation: Scale Height

Deriving the Scale Height

Goal: Derive $H = k_B T / (\mu m_u g)$ from hydrostatic equilibrium + ideal gas law, and compute H for solar system bodies.

Start from:

$$\frac{dP}{dz} = -\rho g \quad \text{and} \quad P = \frac{\rho k_B T}{\mu m_u}$$

Express ρ in terms of P :

$$\rho = \frac{P \mu m_u}{k_B T}$$

Substitute:

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P \equiv -\frac{P}{H}$$

The Atmospheric Scale Height

$$H = \frac{k_B T}{\mu m_u g}$$

Physical interpretation:

- **Higher T** $\rightarrow$ larger H : hotter gas extends further
- **Heavier molecules** (larger μ) $\rightarrow$ smaller H : harder to loft
- **Stronger gravity g** $\rightarrow$ smaller H : more compressed

Key insight: H encodes the competition between thermal energy and gravitational binding.

Worked Example: Earth's Scale Height

$$H_{\oplus} = \frac{k_B T}{\mu m_u g} = \frac{1.381 \times 10^{-23} \times 288}{28.97 \times 1.661 \times 10^{-27} \times 9.81}$$
$$= \frac{3.98 \times 10^{-21}}{4.72 \times 10^{-25}} \approx 8400 \text{ m} \approx 8.4 \text{ km}$$

- Commercial aircraft: 10–12 km (~ 1.2 – $1.4 H$), pressure ~ 0.2 – 0.3 atm
- Top of Everest: 8.8 km ($\sim 1 H$), pressure ~ 0.3 atm
- 99% of mass below ~ 42 km ($\sim 5 H$)

Scale Heights Across the Solar System

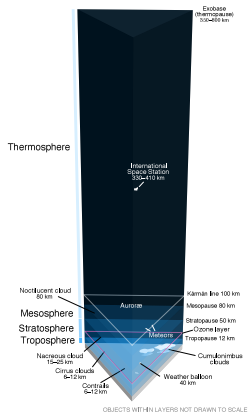
Body	T (K)	μ	g (m s^{-2})	H (km)
Venus	735	43.4	8.87	15.8
Earth	288	29.0	9.81	8.5
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	27
Titan	94	28.6	1.35	21

- Jupiter: large H despite strong gravity, because H_2 is very light ($\mu = 2.2$)
- Titan: large H despite low T , thanks to very weak gravity ($g = 1.35 \text{ m s}^{-2}$)
- Venus: hot but heavy molecules + decent gravity $\rightarrow$ moderate H

4. Vertical Structure



Atmospheric Layers



Credit: Wikimedia, public domain

Layers defined by the sign of $\frac{dT}{dz}$:

1. **Troposphere** (0–12 km): T decreases with z
Heated from below; convective
2. **Stratosphere** (12–50 km): T increases
UV absorbed by ozone layer
3. **Mesosphere** (50–85 km): T decreases
Coldest point at mesopause (~190 K)
4. **Thermosphere** (>85 km): T increases
EUV absorption; $T > 1000$ K but gas is rarefied

The Adiabatic Lapse Rate

In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{dT}{dz} = \frac{g}{c_p}$$

- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8 \text{ K km}^{-1}$
- Observed average: $\sim 6.5 \text{ K km}^{-1}$ (lower due to latent heat from water condensation)
- Physical picture: rising air expands, does work on surroundings, cools

Tropopause height: $\sim 8 \text{ km}$ (poles) to $\sim 17 \text{ km}$ (equator).

Comparative Vertical Structures

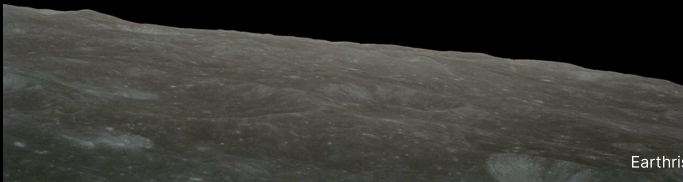
Venus: Massive troposphere (~65 km). No ozone layer → no stratospheric inversion. Cloud deck at 48–70 km.

Mars: Thin troposphere (~40 km), directly into thermosphere. No significant ozone. Dust dominates atmospheric heating.

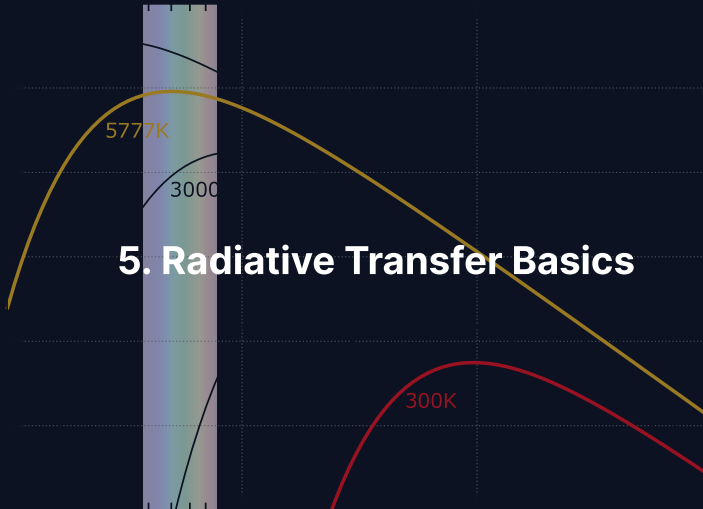
Jupiter: Troposphere extends hundreds of km deep. Stratosphere heated by CH_4 . No solid surface; pressure increases continuously.

Titan: Thick troposphere (~40 km). Stratosphere heated by organic haze. Extended thermosphere reaches ~1500 km (weak gravity).

Break



Earthrise, Apollo 8 (1968). Credit: NASA



Three Interactions: Radiation and Matter

When radiation passes through an atmosphere:

Absorption: Photon energy $\rightarrow$ internal energy (vibrational, rotational)

Key absorbers: CO_2 , H_2O , CH_4 , O_3 , N_2O

Emission: Warm gas radiates at wavelengths where it absorbs (Kirchhoff's law)

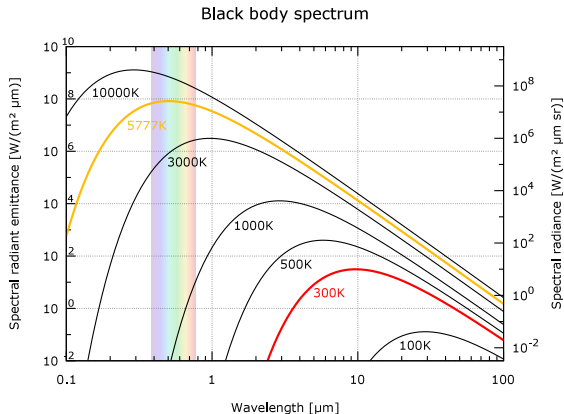
This is how the atmosphere radiates energy to space

Scattering: Photons redirected, not absorbed

Rayleigh ($\propto \lambda^{-4}$): blue sky, red sunsets

Mie (larger particles): less wavelength-dependent

Stellar vs. Planetary Emission



Credit: Wikimedia, public domain

- Sun (~5800 K): peak at ~0.5 μm (visible)
- Planet (~300 K): peak at ~10 μm (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.

Optical Depth

The cumulative absorption along a path through the atmosphere:

$$\tau = \int_0^s \kappa \rho \, ds'$$

- κ = mass absorption coefficient ($\text{m}^2 \text{kg}^{-1}$)
- τ is dimensionless: counts “e-folding lengths” of absorption

Optically thin ($\tau \ll 1$):

Most radiation passes through.

Optically thick ($\tau \gg 1$):

Radiation strongly absorbed; only photons from near the “surface” escape.

The Classical Habitable Zone

Where around a star can a planet maintain liquid water?

- Inner edge: runaway greenhouse limit (water fully evaporates)
- Outer edge: CO₂ condenses into ice clouds (Maximum Greenhouse limit)
- For the Sun: classical HZ is ~0.95–1.67 AU (Kopparapu et al. 2013)
- Assumes an Earth-like atmosphere with CO₂/H₂O feedback
- Does not account for tidal heating, magnetic fields, or greenhouse gas diversity
- A starting point, not the last word

Source: Kasting et al. (1993); Kopparapu et al. (2013)

The Moist Adiabatic Lapse Rate

When a parcel rises and condensation occurs, latent heat release warms it relative to the dry adiabat.

$$\Gamma_{\text{moist}} = \Gamma_{\text{dry}} \cdot \frac{1 + \frac{L_v r_s}{R_d T}}{1 + \frac{L_v^2 r_s}{c_p R_v T^2}}$$

- r_s : saturation mixing ratio; L_v : latent heat of vaporisation
- Earth's lower troposphere: $\Gamma_{\text{moist}} \approx 6 \text{ K/km}$ (compared to 9.8 K/km dry)
- Moist convection is **deeper and more vigorous** than dry; drives thunderstorms and hurricanes
- Latent heat release in rising parcels provides the buoyancy that makes tropical convection reach the tropopause
- Runaway greenhouse reduces Γ_{moist} further, letting more IR escape to

Deriving the Scale Height (Live)

Template for the blackboard derivation.

1. Start: $\frac{dP}{dz} = -\rho g$ (hydrostatic equilibrium)

2. Ideal gas: $P = \frac{\rho k_B T}{\mu m_u}$

3. Rearrange for ρ : $\rho = \frac{\mu m_u P}{k_B T}$

4. Substitute: $\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P$

5. Separate variables and integrate:

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right), \quad H = \frac{k_B T}{\mu m_u g}$$

6. Numerical check: Earth ($T = 288 \text{ K}$, $\mu = 29$, $g = 9.81$) gives $H \approx 8.4 \text{ km}$

Transmission Spectroscopy: Preview

During a transit, starlight passes through the planet's **limb**.

- Absorbing molecules imprint dark features in the transmission spectrum
- Path length $\sim 2\sqrt{2R_p H}$ through a layer of scale height H
- Signal: $\Delta(R_p/R_\star)^2 \sim 2R_p H/R_\star^2$
- Bigger signal for puffy atmospheres (large H) around small stars
- JWST has detected H_2O , CO_2 , CH_4 , SO_2 in multiple exoplanet atmospheres
- Full treatment in Lecture 13 (Exoplanet Atmospheres)

Source: Seager & Sasselov (2000); Kreidberg et al. (2014)

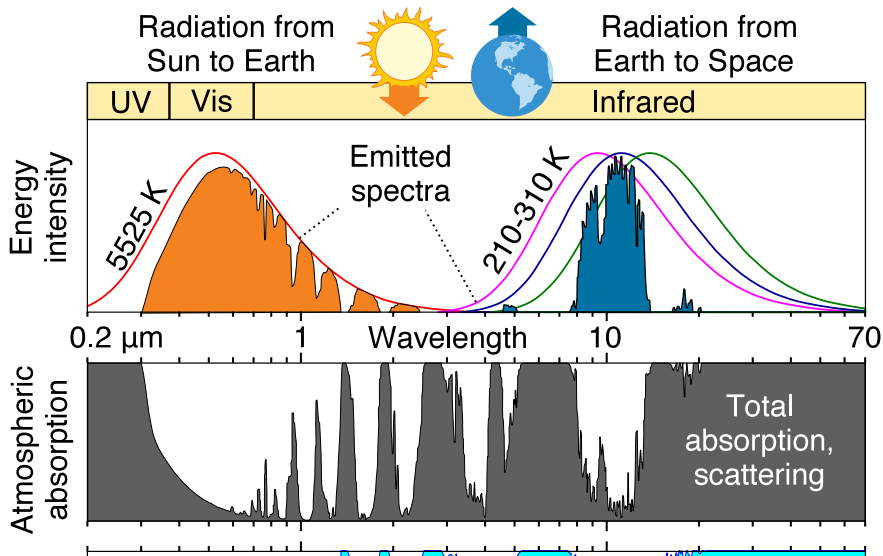
The Beer–Lambert Law

For a beam of initial intensity I_0 through a medium of optical depth τ :

$$I = I_0 e^{-\tau}$$

- Intensity drops by factor e per unit optical depth
- **Atmospheric photosphere:** at $\tau \approx 1$, photons escape to space
- Below $\tau = 1$: photons reabsorbed before escaping
- Above $\tau = 1$: photons escape freely
- Temperature at the $\tau \approx 1$ level sets T_{eff}

Earth's Atmospheric Absorption Spectrum



6. Energy Balance & Greenhouse Effect



Radiative-Convective Equilibrium

A realistic atmosphere combines two regimes:

- **Troposphere:** strong greenhouse absorption → unstable to convection
 - Temperature follows the adiabatic lapse rate
 - Most of the atmospheric mass
 - Clouds and weather occur here
- **Stratosphere:** optically thin → radiative equilibrium
 - Temperature set by the balance of absorbed vs. emitted radiation
 - Stable against convection
 - On Earth, warmed by ozone UV absorption

The boundary (the tropopause) occurs where the radiative gradient becomes less steep than the adiabat.

Manabe and Strickler (1964)

The first realistic 1D radiative-convective model of Earth's atmosphere.

- Solved for the vertical temperature profile with absorbing gases (H_2O , CO_2 , O_3)
- Reproduced the surface temperature and the tropopause self-consistently
- Quantified the effect of doubling CO_2 : ~ 2 K warming (first “climate sensitivity” estimate)
- Foundation of modern climate modelling
- Manabe received the 2021 Nobel Prize in Physics for this work

Source: Manabe & Strickler (1964), *J. Atmos. Sci.*

Planetary Energy Balance

Absorbed stellar power = emitted thermal radiation:

$$(1 - A) \frac{L_{\star}}{4\pi d^2} \pi R_p^2 = 4\pi R_p^2 \sigma T_{\text{eff}}^4$$

- A = Bond albedo (fraction of light reflected)
- πR_p^2 = cross-sectional area (intercepts light)
- $4\pi R_p^2$ = total surface area (emits isotropically)

Solving:

$$T_{\text{eff}} = \left[\frac{(1 - A) L_{\star}}{16\pi\sigma d^2} \right]^{1/4}$$

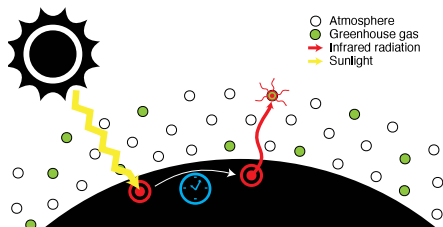
Effective vs. Surface Temperature

Body	A	d (AU)	T_{eff} (K)	T_{surf} (K)	ΔT (K)
Venus	0.77	0.72	227	735	+508
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	210	~0
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\text{surf}} - T_{\text{eff}}$ reveals the **greenhouse effect**.
Venus: 508 K. Earth: 33 K. Mars: ~0 K (too thin).

The Greenhouse Mechanism



Credit: Wikimedia, public domain

1. Sunlight (visible, $\sim 0.5 \mu\text{m}$) reaches the surface
2. Surface emits thermal IR ($\sim 10\text{--}15 \mu\text{m}$)
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

Result: $T_{\text{surf}} > T_{\text{eff}}$

One-Layer Greenhouse Model: Setup

Single isothermal layer, emissivity ε in IR, transparent at visible:

Atmospheric balance:

$$\varepsilon \sigma T_s^4 = 2 \varepsilon \sigma T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{T_s^4}{2}$$

Surface balance:

$$(1 - A) \frac{F_\star}{4} + \varepsilon \sigma T_a^4 = \sigma T_s^4$$

Top-of-atmosphere:

$$(1 - A) \frac{F_\star}{4} = \sigma T_{\text{eff}}^4$$

One-Layer Greenhouse Model: Result

Combining the three balance equations:

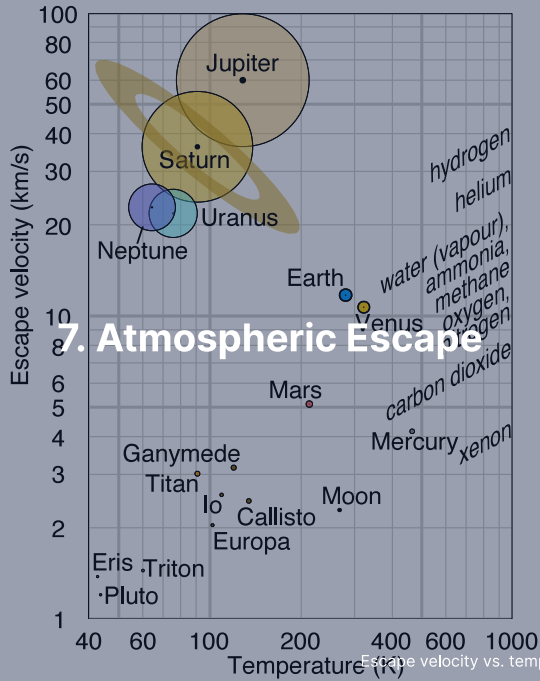
$$T_s = T_{\text{eff}} \left(\frac{2}{2 - \varepsilon} \right)^{1/4}$$

$$\varepsilon = 0 \text{ (no greenhouse): } T_s = T_{\text{eff}}$$

$$\varepsilon = 1 \text{ (perfect absorber): } T_s = 2^{1/4} T_{\text{eff}} \approx 1.19 T_{\text{eff}}$$

For Earth: $T_s \approx 1.19 \times 255 \approx 303$ K. Reasonable first estimate.

Next lecture: What happens when T_s rises so high that outgoing radiation saturates? → *Runaway greenhouse* (Lecture 6).



Thermal (Jeans) Escape

Molecules have a Maxwell–Boltzmann velocity distribution. The high-velocity tail extends beyond escape speed.

The **Jeans escape parameter** at the exobase:

$$\lambda_J = \frac{GM_p m}{k_B T_{\text{exo}} r_{\text{exo}}} = \frac{v_{\text{esc}}^2}{v_{\text{th}}^2}$$

- $\lambda_J \gg 1$: escape negligible (tail too depleted)
- $\lambda_J \lesssim 2\text{--}3$: rapid atmospheric erosion
- Escape flux: $\Phi_J \propto (1 + \lambda_J) e^{-\lambda_J}$, exponentially sensitive

Jeans Parameter for Earth and Mars

Exobase temperatures: 1000 K (Earth), 300 K (Mars)

Species	m (u)	λ_j (Earth)	λ_j (Mars)
H	1	7.5	5.4
H ₂	2	15	11
He	4	30	22
N ₂	28	210	150
CO ₂	44	330	240

Heavy species (N₂, CO₂): λ_j so large that Jeans escape is negligible.
Atomic H: moderate $\lambda_j \rightarrow$ both Earth and Mars lose hydrogen to space.

Hydrodynamic Escape

When stellar EUV heating is intense, escape transitions from molecule-by-molecule to a bulk **atmospheric wind**:

- Most important in the first ~100 Myr (young stars: 10–100× more EUV)
- Can strip H₂-rich primary atmospheres from planets up to several $M_{\oplus}$
- Outflowing H can **drag along heavier species** (He, C, N, O)
- Leading explanation for the exoplanet **radius valley** (deficit at ~1.5–2 $R_{\oplus}$; Lecture 13)

Stripping Primary Atmospheres

Young stars are highly active in EUV and X-ray; primordial H/He envelopes can evaporate.

- EUV flux is $\sim 100\times$ higher in the first 100 Myr than today
- Mass loss rate for hydrodynamic regime: $\dot{M} \propto F_{\text{EUV}} R_p^3 / (GM_p)$
- For sub-Neptunes ($M_p \lesssim 10 M_\oplus$), atmospheres can be fully stripped in a few hundred Myr
- For Neptune-mass planets and above, envelopes are retained
- Produces the “evaporation valley” at $\sim 1.8 R_\oplus$ in Kepler data

Source: Owen & Wu (2013); Lopez & Fortney (2013); Fulton et al. (2017)

Energy-Limited Escape Rate

A convenient estimate of mass loss:

$$\dot{M}_{\text{EL}} = \eta \frac{\pi F_{\text{EUV}} R_p^3}{G M_p}$$

- F_{EUV} : incident EUV flux
- η : heating efficiency, typically 0.1–0.2
- Scales as R_p^3 : large, fluffy atmospheres lose mass fastest
- Scales inversely with M_p : low gravity eases escape
- Valid when recombination is negligible (“energy-limited regime”)

Source: Watson et al. (1981); Owen (2019)

Non-Thermal Escape Mechanisms

Sputtering: Solar wind ions impact upper atmosphere, eject molecules.
Significant for Mars (no global magnetic field).

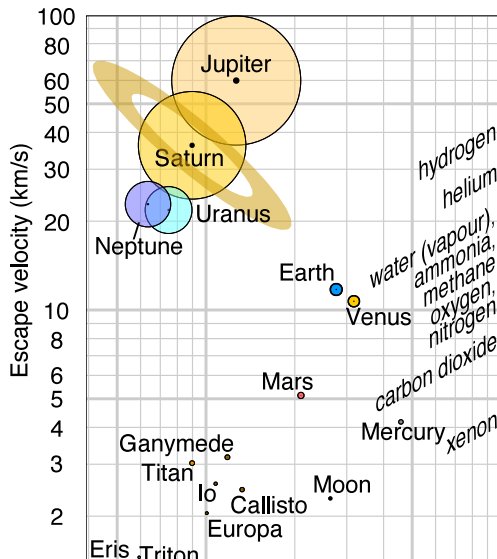
Photochemical: UV dissociates $\text{H}_2\text{O} \rightarrow \text{H} + \text{OH}$; fast H escapes.
Key for Venus and Mars water loss.

Ion pickup: Atmospheric atoms ionised by UV, swept away by solar wind B -field.
Effective at unmagnetised planets.

Impact erosion: Large impacts can eject substantial atmospheric mass in a single event.

MAVEN at Mars: present loss rate $\sim 100 \text{ g s}^{-1}$ (O^+), dominated by solar wind stripping.

Atmospheric Retention: v_{esc} vs. T Diagram

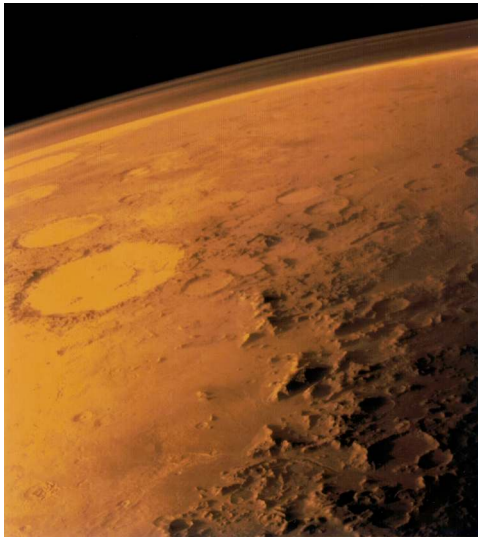


Rule of thumb for long-term retention:

$$v_{\text{esc}} \geq 6 v_{\text{th}}$$

- Gas giants: retain **everything**
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold $\rightarrow$ retains N_2 despite low v_{esc}

Mars: A Lost Atmosphere



Credit: NASA/JPL (Viking 1, 1976)

- Present surface pressure: only 6 mbar
- Too thin for liquid water or significant greenhouse warming
- Geological evidence: ancient rivers, lakes, possibly ocean
- Mars had a much thicker atmosphere >4 Gyr ago
- Dynamo loss (~4.1 Ga, Lecture 4) → unshielded solar wind erosion



8. Recent Advances & Outlook

Recent Advances

- **JWST / TRAPPIST-1 b:** Thermal emission consistent with bare rock, no significant atmosphere. Inner rocky planets around active M dwarfs may be stripped.
- **JWST / TRAPPIST-1 c:** Similar result, reinforcing theoretical predictions of enhanced atmospheric escape around low-mass stars.
- **MAVEN at Mars:** Quantified loss rates for multiple species. Ion escape (solar wind stripping) dominates over Jeans escape. Integrated losses can account for a substantial fraction of Mars's early atmosphere.

Central theme: atmospheric retention depends on mass, temperature, stellar radiation, magnetic shielding, and geological activity.

Impact Erosion of Atmospheres

Giant impacts can eject substantial atmospheric mass in a single event.

- Shockwaves from a >km-sized impactor accelerate upper-atmospheric gas above escape velocity
- Efficiency depends on impact angle, velocity, and atmosphere mass
- A grazing giant impact can eject the entire atmosphere above the impact horizon
- Key process during the first ~100 Myr of solar system history
- May explain why Earth's early atmosphere was lost despite adequate gravity
- Signature: noble gas isotope ratios ($^{36}\text{Ar}/^{38}\text{Ar}$) record impact-driven loss

Source: Schlichting & Mukhopadhyay (2018); Genda & Abe (2005)

Outgassing vs. Escape: A Dynamic Balance

An atmosphere is not a static inheritance; it is a reservoir in flow.

- **Supply:** volcanic outgassing, impact delivery, biological fluxes
- **Loss:** thermal escape, non-thermal escape, chemical weathering, sequestration in oceans/crust
- The **steady state** depends on the balance of these fluxes
- Earth's active volcanism continually replaces CO_2 on the carbonate-silicate timescale (~ 0.5 Myr)
- Mars lost its active volcanism and dynamo, so escape was not replenished
- Venus has active (perhaps episodic) volcanism, but no oceans to absorb CO_2

Atmospheric longevity is as much about **replacement** as about initial inventory.

MAVEN at Mars: Quantifying Escape

NASA's MAVEN (2014–) has measured Mars' atmospheric loss in situ.

- Present-day total escape rate: ~1–2 kg/s (all species)
- Dominant loss channel: photochemical escape of oxygen
- Enhanced during solar storms (~10× increase)
- Integrated loss over 4 Gyr: consistent with 0.5–0.8 bar of CO₂ lost
- Combined with Jeans escape of hydrogen: ~15–25 m global equivalent layer of water lost

Source: Jakosky et al. (2018), *Icarus*; Lillis et al. (2017)

JWST and the Fate of Rocky Atmospheres

JWST has begun to directly test whether rocky planets around M dwarfs retain atmospheres.

- TRAPPIST-1 b: MIRI thermal emission consistent with a bare rock (dayside 508 K)
- TRAPPIST-1 c: MIRI shows low heat redistribution $\Rightarrow$ thin or no atmosphere
- LHS 3844 b: similar result using Spitzer data
- Implications: inner rocky planets around active M dwarfs lose their atmospheres
- Open question: whether **outer** TRAPPIST-1 planets retain atmospheres
- JWST will target TRAPPIST-1 e, f during Cycle 3–4

Source: Greene et al. (2023); Zieba et al. (2023), *Nature*

Runaway Greenhouse: Preview for Lecture 6

What happens when a water-rich atmosphere gets too hot?

- More water vapour → stronger greenhouse → hotter surface
- Positive feedback: hotter surface → more evaporation → more greenhouse
- At $\sim 360 \text{ W/m}^2$ absorbed, the **Simpson-Nakajima limit** is reached
- Beyond this, oceans fully evaporate and the runaway becomes unstoppable
- Hydrogen then escapes hydrodynamically, leaving a dry CO_2 atmosphere
- Venus' modern state is the end point of this process (Nakajima 1992; Goldblatt 2012)

Full treatment in **Lecture 6: Atmospheres II**.

Summary

1. Three atmosphere types: **primary** (captured H_2/He), **secondary** (outgassed), **tertiary** (biologically modified)
2. Hydrostatic equilibrium: $\frac{dP}{dz} = -\rho g$; leads to exponential pressure decay
3. Scale height: $H = k_B T / (\mu m_u g)$; encodes thermal energy vs. gravity
4. Vertical structure: troposphere, stratosphere, mesosphere, thermosphere
5. Beer–Lambert law: $I = I_0 e^{-\tau}$; $\tau \approx 1$ defines the emission level
6. Energy balance: $T_{\text{eff}} \propto [(1 - A)L_\star / d^2]^{1/4}$; greenhouse raises T_s above T_{eff}
7. Escape: Jeans (λ_j), hydrodynamic, non-thermal; retention requires $v_{\text{esc}} \gtrsim 6 v_{\text{th}}$

Questions?

Next: **Lecture 6. Atmospheres II: Clouds, Weather, & Climate**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience